Multiple IV Bolus Practice 2

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

The elimination half-life of an antibiotic is 3 hours and the apparent volume of distribution is 20% of the body weight. The therapeutic window for this drug is from 2 to 10 mcg/mL. Adverse toxicity is often observed at drug concentrations above 15 mcg/mL. The drug will be given by multiple IV bolus injections.

a Calculate the dose for an adult male patient (68 years old, 82 kg) with normal renal function to be given every 8 hours. (Hint, I started by assigning 10 mg/L as the maximum concentration at steady state).

b Calculate the maximum steady-state plasma concentration. (The regimen from part a: 138.16 mg every 8 hours to an 82-kg patient.)

c Calculate the minimum steady-state plasma concentration. (The regimen from part a gives a steady-state peak of 10 mg/L every 8 hours.)

d Calculate the average steady-state plasma concentration. (The regimen from part a: 138.16 mg every 8 hours to an 82-kg patient.)

e Does your regimen provide the desired peak and trough concentrations? If not, how would you adjust to achieve the desired concentrations? (The regimen from part a: 138.16 mg every 8 hours, giving a steady-state peak of 10 mg/L and a trough of 1.576 mg/L.)

A. Yes: both the peak and the trough stay below the 15 mcg/mL toxic level
B. No: the trough is below 2 mg/L; shorten the interval from 8 to 6 hours
C. No: the peak is too high; lower the dose and keep the 8-hour interval
D. No: the trough is too low; raise the dose and keep the 8-hour interval

e With the dosing interval decreased from 8 to 6 hours, what dose every 6 hours gives a steady-state peak of 10 mg/L? (82-kg patient, V_D 16.4 L.)

e On the new proposed regimen, 123 mg every 6 hours with an expected steady-state peak of 10 mg/L, what is the expected steady-state trough?